Competent
86/90
96%

Module #	671101000-KM-07:
NQF Level	level 4,
Notional hours	310
Credit(s)	Cr 31
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS

FORMATIVE ASSESSMENT

About the Formative Assessment	
Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

Special arrangements for assessment

Place	East C Electrical
Language	English
Resources	
Barriers	N/A

People to be involved with assessment

Learner:	Sela Phiri
Trainer:	Manager:
Facilitator	Miss R. NYAKALA Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Sela Phiri

Learner's signature: S. Phiri Date:

Facilitator Name:

Facilitator's signature: Date:

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Stla	Signature S.
Facilitator Name: Miss R. Nyakala	Signature:

LEARNING UNIT 1: KM-07-KT01: Concepts, theories and principles of supply Systems (20%)	
Learning Outcome 1:	KT0101 Theories and concepts of alternating current
Learning Outcome 2:	KT0102 Fundamental principles of alternating current
Learning Outcome 3:	KT0103 Alternating current generation (distribution systems theory)
Learning Outcome 4:	KT0104 Characteristics and calculations from alternating current waveforms
Learning Outcome 5:	KT0105 Theories, concepts and principles of direct current sources
Learning Outcome 6:	KT0106 Sources of direct current
Learning Outcome 7:	KT0107 Calculations of direct and alternating current circuits
Learning Outcome 8:	KT0108 Construction and operating principles of direct current generators
Learning Outcome 9:	KT0109 Characteristics, sources and generation of renewable energy
Learning Outcome 10:	KT0110 Relevant legislative requirements

Task	Questions Description	Marks
1.	a) Explain the generation and differences between single and three phase alternating current by using wave forms and vector diagrams.	4
	b) Explain by drawing wave form diagrams, the differences between line and phase values.	4
	c) Explain the generation principles of direct current.	4
	d) Calculate power and energy in direct and alternating current circuits	4
	e) Explain with the aid of drawings the operating principles of direct current generators.	4

(d) Power & Energy in Circuit

DC Circuits

Power = $P = V \cdot I$

Energy = $E = P \cdot t$

AC Circuits

instantaneous power

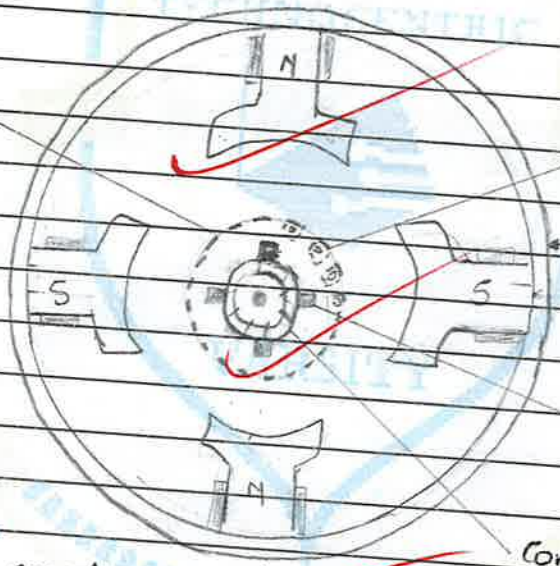
$p(t) = v(t) \cdot i(t)$

Average power

$P = V_{rms} \cdot I_{rms} \cdot \cos \phi$ (where ϕ = phase angle)

Energy = $E = P \cdot t$

e.



A DC generator operates on the principle of electromagnetic induction i.e When the magnetic flux linking a conductor changes ~~A DC gene~~ an EMF induced in the conductor.

A DC generator has a field winding and an armature windings.

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(b)

Primary cell	Secondary cells
Advantages	Advantages
*lower initial cost : Generally Cheaper to manufacture than Secondary	*Rechargeable : Can be reused hundreds or thousands of times reducing waste
*long shelf life : Can be sorted for extended periods without significant loss of ~~ex~~ Charge	*lower long-term cost : more economical for frequently used devices
Disadvantages	Disadvantages
*Non-rechargeable : Once depleted they are waste	* Higher initial cost : more expensive to purchase initially
* Higher long-term cost : For Frequently used devices the cost of replacement can be high	* Self-discharge : Tend to lose charge over time even when not in use
*Environmental impact : Disposal of used batteries contributes to waste	a limited lifespan : Although rechargeable, they eventually degrade and need replacement

c Voltmeter

(c)

Instrument	How they are used	The purpose of the test
1. Voltmeter	*A voltmeter is connected in parallel across the battery terminals	*TO measure the open-circuit voltage (voltage when no current is flowing) or voltage when the battery is supplying current to a load
2. Hydrometer	*Is used specifically for lead acid batteries	* Determines the charge level of each cell

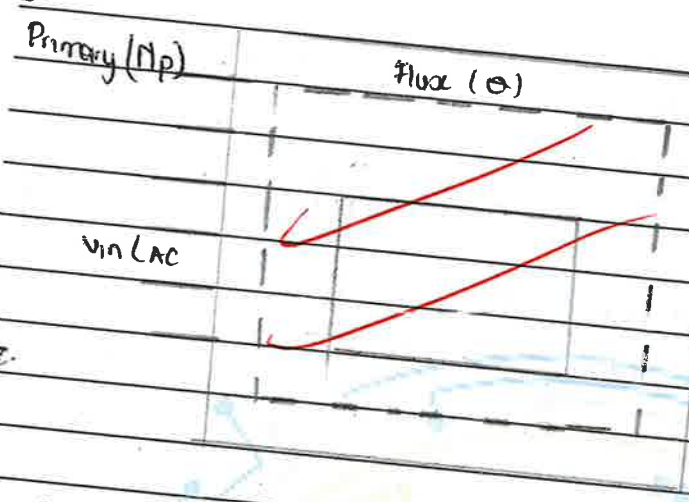
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LEARNING UNIT 3: KM-07-KT03: Transformers (20%)	
Learning Outcome 1:	KT0301 Theories of single phase wound transformers
Learning Outcome 2:	KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications.
Learning Outcome 3:	KT0303 Fundamentals of transformer construction
Learning Outcome 4:	KT0704 Transformer cooling systems
Learning Outcome 5:	KT0305 Principles of single phase transformer operation
Learning Outcome 6:	KT0306 Principles of single phase auto-transformer operation
Learning Outcome 7:	KT0307 Transformer losses
Learning Outcome 8:	KT0308 Formulas and calculations on input and output of transformers
Learning Outcome 9:	KT0709 Maintenance requirements of transformers
Learning Outcome 10:	KT0310 Types of three phase transformers and their applications
Learning Outcome 11:	KT0311 Construction of three phase transformers including open core, closed core and shell or divided core
Learning Outcome 12:	KT0312 Principles of 3 phase transformer operation and configuration
Learning Outcome 13:	KT0313 Principles of 3 phase auto transformer operation and configuration
Learning Outcome 14:	KT0314 Transformer protection and phasing

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S

b.



c.

(c) Copper losses -

Cause

- Current flowing through the winding resistance
- According to ohm's law, power loss is : $P = I^2R$

Effect

- Produces heat in the windings
- Causes temperature rise
- Increase with load current

* Core losses (Iron losses)

Cause

- Molecular friction inside the core material due to alternating magnetic flux

Effect

d. Turns ratio

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

Current Ratio

$$\frac{I_1}{I_2} = \frac{N_2}{N_1}$$

V = voltage
I = Current
N = number of turns

If $N_2 > N_1$ → Step-up transformer

If $N_2 < N_1$ → Step down transformer

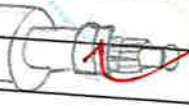
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LEARNING UNIT 4:
KM-07-KT04: Types of cables and applications (20%)

Learning Outcome 1:	KT0401 Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables.
Learning Outcome 2:	KT0402 Cable materials and their functions and characteristics
Learning Outcome 3:	KT0403 Identification of cable characteristics and properties
Learning Outcome 4:	KT0404 Applications of various types of cable
Learning Outcome 5:	KT0405 Installation methods and safe use of cables
Learning Outcome 6:	KT0406 Safe transport and storage of cables

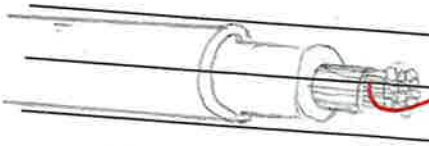
Task	Questions Description	Marks
4	a) Describe, with the aid of sketches, the different types of cables and their construction, and explain the function of the various materials used	4
	b) Describe the factors effecting the effeciency of cables	3
	c) Describe the precautions required when transporting and storing cables	3

• XLPE Insulated Cable



• Copper / Aluminium Conductor
• XLPE insulation
• Screen
• Outer PVC Sheath

Armoral Cable (SWA)



• Copper / Aluminium Conductor
• PVC / XLPE insulation
• Steel Wire Armour
• Outer PVC Sheath

Fuction of Materials
• Armour (steel wires) - protects against mechanical damage and can act as earth path

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LEARNING UNIT 5:

KM-07-KT05: Switchgear and control gear (20%)

Learning Outcome 1:	KT0501 Principles of operation of switchgear and control gear
Learning Outcome 2:	KT0502 Components of switchgear and control gear systems and the application thereof.
Learning Outcome 3:	KT0503 Electrical drawings
Learning Outcome 4:	KT0504 Introduction to Programmable Logic Controllers (PLC's)
Learning Outcome 5:	KT0505 Introduction to Soft starters
Learning Outcome 6:	KT0506 Introduction to Variable speed drives

Task	Questions Description	Marks
5	a) Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and operating mechanisms	4
	b) Describe the basic principles of operation for programmable logic controllers	3
	c) Describe the basic principles of operation for variable speed drives	3

Disconnectors

Contr

	Construction	operating Principle	Contacts	Mechanism
• Disconnectors	• Fixed and moving metal contacts	• works purely mechanically	• Heavy-duty contacts designed for visibility and safety	• Manual lever • motor driven linkage

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Rotor Resistance Control (Slip Ring motors)	• External resistance is added to the rotor circuit	• Increase rotor resistance → increase slip → reduces speed • Decrease resistance → increase speed

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b.

Feature	Single-Phase motor	Three-Phase motor
Self-Starting	No- requires auxiliary windings or capacitors to start	Yes. Naturally generates a rotating magnetic field
Torque	Pulsating torque, less stable	Smooth, continuous and high torque
Efficiency	lower (typically 70-85%)	Higher (typically 85-96%)
Power Range	Generally up to 5HP	Fractional to thousands of HP

C. Testing alternating current (AC) motors involves a combination of static (OFFline) test to check physical integrity and dynamic (online) tests are to verify performance under load. These procedures ensure safety, efficiency, and early detection of potential failures like insulation breakdown or mechanical Were

- Insulation Resistance (IR) and megger testing
A megohmmeter (megger applies a high DC voltage (typically 500V to 1000V) between the motor windings and the ground (Frame)

d. Speed (RPM) $Ns = \frac{120f}{p}$

Speed Full load Amps (3Ø $I = \frac{HP \times 746}{\sqrt{3} \times 735 \times h \times PF}$

out power (Kw) $P_{out} = \frac{T \times RPM}{9550}$ (Torque in Nm)

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ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT		
Learner's Total Mark:	**Requirements met** ☐	**Requirements not met** ☐
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR
I ..... (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

S

Application for Assessment

First Attempt □ Second Attempt □ Third Attempt□ RPL □

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments 4. Conduct observation in workplace where applicable 5. Complete learner toolkit 6. Complete assessor toolkit

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